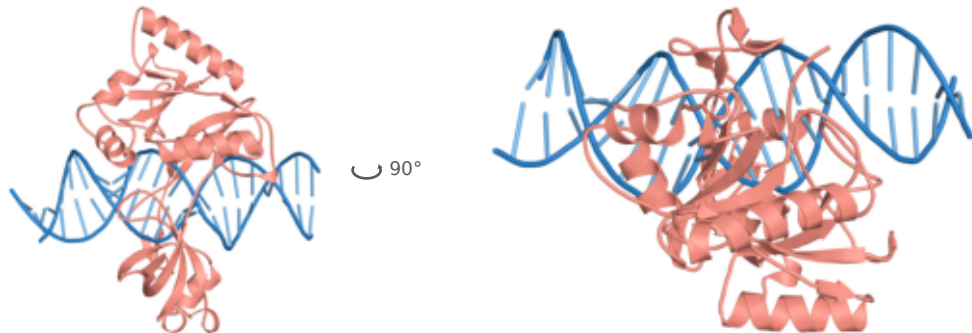
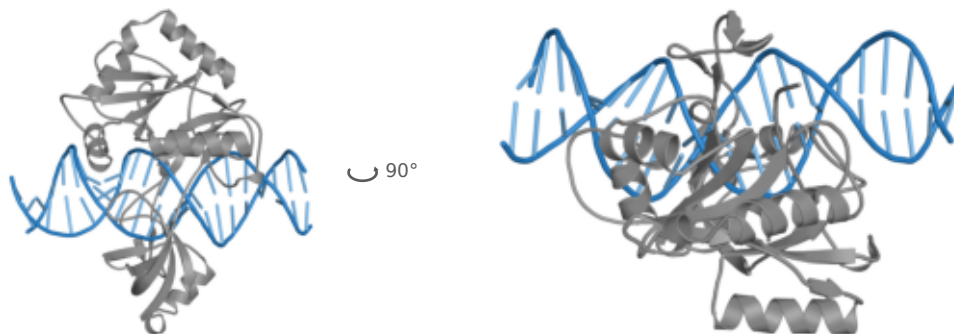


A

ORF55 + DNA (AF3)



Ligase + DNA (2Q2U)

*Adenylylation active site***B**

K27-K25

D29-D27

R32-R30

E67-E66

F98-F99

E161-E161

K186-R186 (+2)

K188-K188

*Nick recognition*

R42-R41

R176-R176



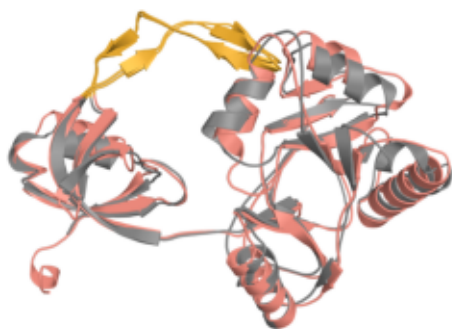
■ ORF55 (DruSM1 phage)
■ DNA ligase (Chlorella virus)
■ OB-domain latch

Substrate & clamp geometry

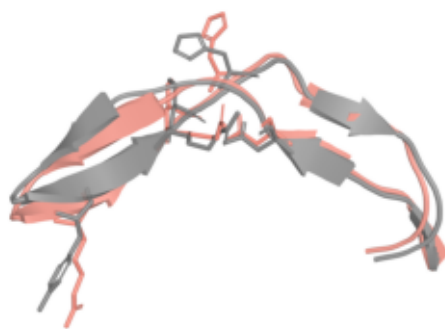
F75-C78 (-2)

R285-R288

F286-F289

*OB-domain latch (residues 202-231)*

the latch (gold) in the two superimposed folds



magnified — ORF55 (coral) over the ligase (grey)